

Name of the student:

Sch. No. 56

MAULANA AZAD NATIONAL INSTITUTE OF TECHNOLOGY, BHOPAL
Department of Electrical Engineering

End Term Examination, Nov. 2023

Course: B.TECH

Semester: III

Branch: Electrical Engg.

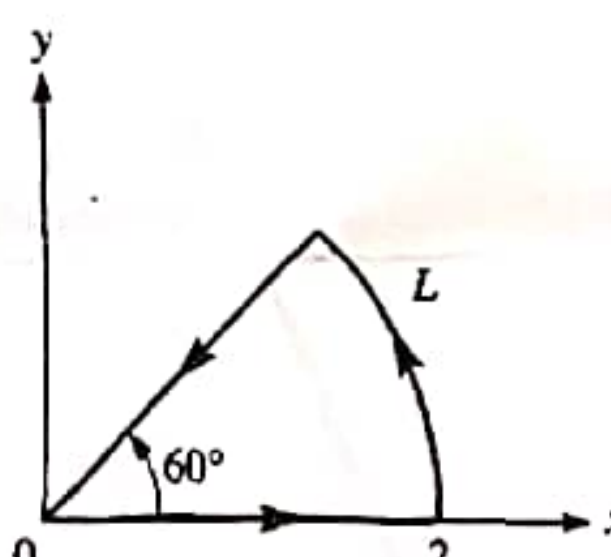
Subject Name: Electromagnetic Fields & Materials

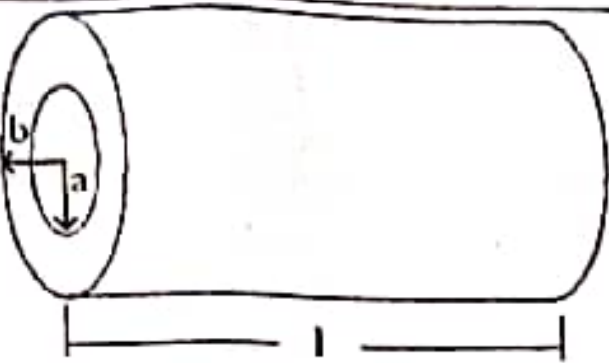
Subject code: EE-211

Time: 03 Hrs.

Max. Marks: 50

Note: Answer all questions. Assume missing data if necessary.

Q. No.	Questions	Marks	
I.	a) Derive the generalized relation for energy stored in a electric fields of continuous charge distribution.	05	CO2
	b) Calculate the circulation of vector $A = \rho \cos \phi \hat{\rho} + z \sin \phi \hat{z}$ around the edge L of the wedge defined by $0 \leq \rho \leq 2$, $0 \leq \phi \leq 60^\circ$, $z = 0$ and shown in figure below.	04	CO1
			
II.	a) Drive the relation $E = -\nabla V$, where E & V are electric field and electric potential respectively.	04	CO1
	b) Derive Poisson's equation in rectangular coordinates and write poisson's equation in general coordinate system. Using Laplace's equation, solve the potential function between the conductors of coaxial cable, if V_0 is the potential difference between them and radius of inner and outer conductors are a & b respectively.	05	CO2
III	a) Region-1 is $z \leq 0$ is a dielectric for which $\epsilon_r = 2$, while region-2, $z > 0$, is a free space. Given $D_1 = 3\hat{x} - 4\hat{y} + 6\hat{z}$. Find D_2 & E_2 .	04	CO1
	b) Determine the resistance of the insulation in a length ' l ' of a coaxial cable as shown in the fig. conductivity of insulation is ' σ '	04	CO2

			
IV	a) Derive the Maxwell's second equation based on Ampere's law	04	CO3
	b) Derive the boundary relation of tangential and normal component of magnetic fields at the boundary of two different media.	04	CO3
V.	a) A filamentary current of 10 A is directed in from infinity to the origin on the positive X-axis, and then back out to infinite along the positive y-axis. Use Biot-Savart law to find H at $P(0\ 0\ 1)$.	04	CO3
	b) For a uniform plane wave determine the ratio of magnitude of Electric field to magnetic field in air medium.	04	CO4
VI	a) Give the comparison of Copper and Aluminum as conductors for power transmission lines.	04	CO5
	b) Discuss the materials used of making permanent magnets.	04	CO5

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